DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: First Year B. Tech. | **Branch: Group A / Group B**

Subject Name: Engineering Mechanics | **Subject Code: BTES203**

Max Marks: 60 | **Date:19-01-24** | **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Define characteristics of system of forces.	**Remember**	**6**
B)	The forces 20 N, 30 N, 40 N, 50 N and 60 N are acting at one of the angular points of a regular hexagon, towards the other five angular points, taken in order. Find the magnitude and direction of the resultant force.	**CO2**	**6**

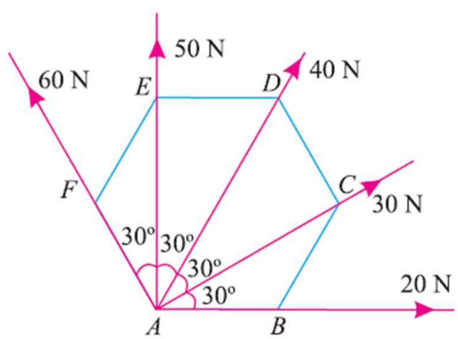

Fig. 1

C)	A uniform wheel of 600 mm diameter, weighing 5 kN rests against a rigid rectangular block of 150 mm height as shown in Fig. 2.	**CO2**	**6**

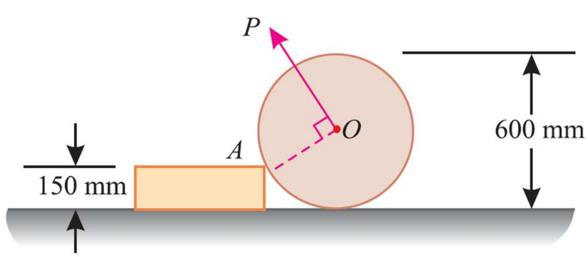

Fig. 2

Find the least pull, through the centre of the wheel, required just to turn the wheel over the corner A of the block. Also find the reaction on the block. Take all the surfaces to be smooth.

Q.2 Solve Any Two of the following. **12**

A) a) What are the characteristics of a couple? **CO1** **6**

b) Fig. 3 shows a crank-lever ABC with a tension spring (T). The lever weighs 0.2 N/mm. Determine the tension developed in the spring, when a load of 100 N is applied at A.

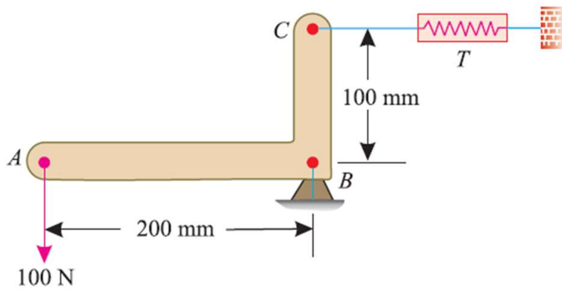

Fig. 3

B) A light string ABCDE whose extremity A is fixed, has weights W1 and W2 attached to it at B and C. It passes round a small smooth peg at D carrying a weight of 300 N at the free end E as shown in Fig. 4 **CO2** **6**

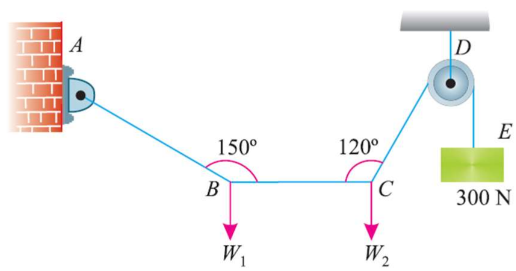

Fig. 4

If in the equilibrium position, BC is horizontal and AB and CD make 150° and 120° with BC, find

(i) Tensions in the portion AB, BC and CD of the string and

(ii) Magnitudes of W_1 and W_2

C) A semicircular area is removed from a trapezium as shown in Fig. 5 (dimensions in mm). **CO3** **6**

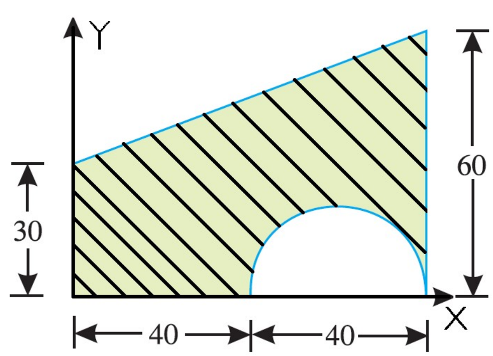

Fig. 5

Determine the centroid of the remaining area (shown hatched).

Q. 3 Solve Any Two of the following. **12**

A) i. Define Following **CO1** **6**

a. Static Friction

b. Dynamic Friction

c. Coefficient of Friction

ii. Find the horizontal force required to drag a body of weight 100 N along a horizontal plane. If the plane, when gradually raised up to 15°, the body will begin to slide.

B) Fig. 6 shows a Warren girder consisting of seven members each of 3 m length freely supported at its end points. **CO2** **6**

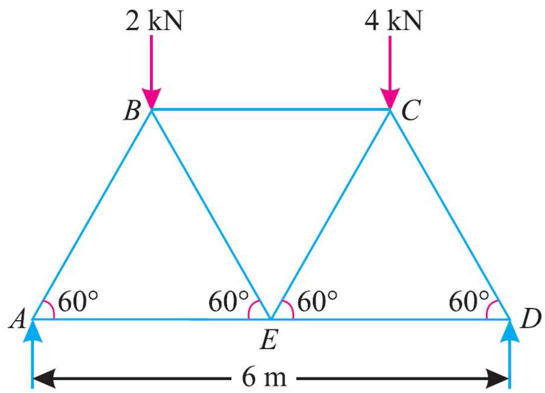

Fig. 6

The girder is loaded at B and C as shown. Find the forces in all the members of the girder, indicating whether the force is compressive or tensile.

C) A simply supported beam AB of span 5 m is loaded as shown in Fig. 7. Using principle of virtual work, find the reactions at A and B. **CO5** **6**

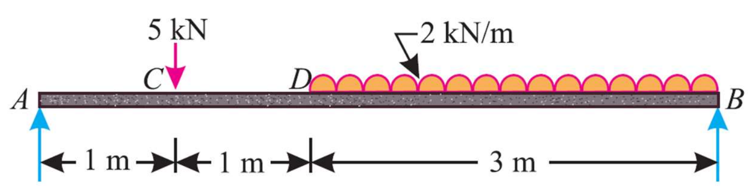

Fig. 7

Q.4 Solve Any Two of the following. **12**

A) i. Define D'Alembert's Principle and write down its equation. **CO1** **6**

ii. Define Newton's First, Second and Third Law of Motion.

B) Two electric trains A and B leave the same station on parallel lines. The train A starts from rest with a uniform acceleration of 0.2 m/s2 and attains a speed of 45 kmph., which is maintained constant afterwards. The train B leaves 1 minute after with a uniform acceleration of 0.4 m/s2 to attain a maximum **CO4** **6**

speed of 72 kmph., which is maintained constant afterwards. When will the train B overtake the train A?

C)	A car moves along a straight line whose equation of motion is given by $s = 12t + 3t^2 - 2t^3$, where (s) is in metres and (t) is in seconds. Calculate (i) velocity and acceleration at start, and (ii) acceleration, when the velocity is zero.	**CO4**	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	Derive equation of the path of a projectile and write equation of time of flight, horizontal range, and maximum height of a projectile.	**CO5**	**6**
B)	A flywheel is making 180 r.p.m. and after 20 sec it is running at 120 r.p.m. How many revolutions will it make and what time will elapse before it stops, if the retardation is uniform?	**CO4**	**6**
C)	A bullet of mass 20 g is fired horizontally with a velocity of 300 m/s, from a gun carried in a carriage, which together with the gun has mass of 100 kg. The resistance to sliding of the carriage over the ice on which it rests is 20 N. Find a) velocity, with which the gun will recoil, b) distance, in which it comes to rest, and c) time taken to do so.	**CO5**	**6**

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